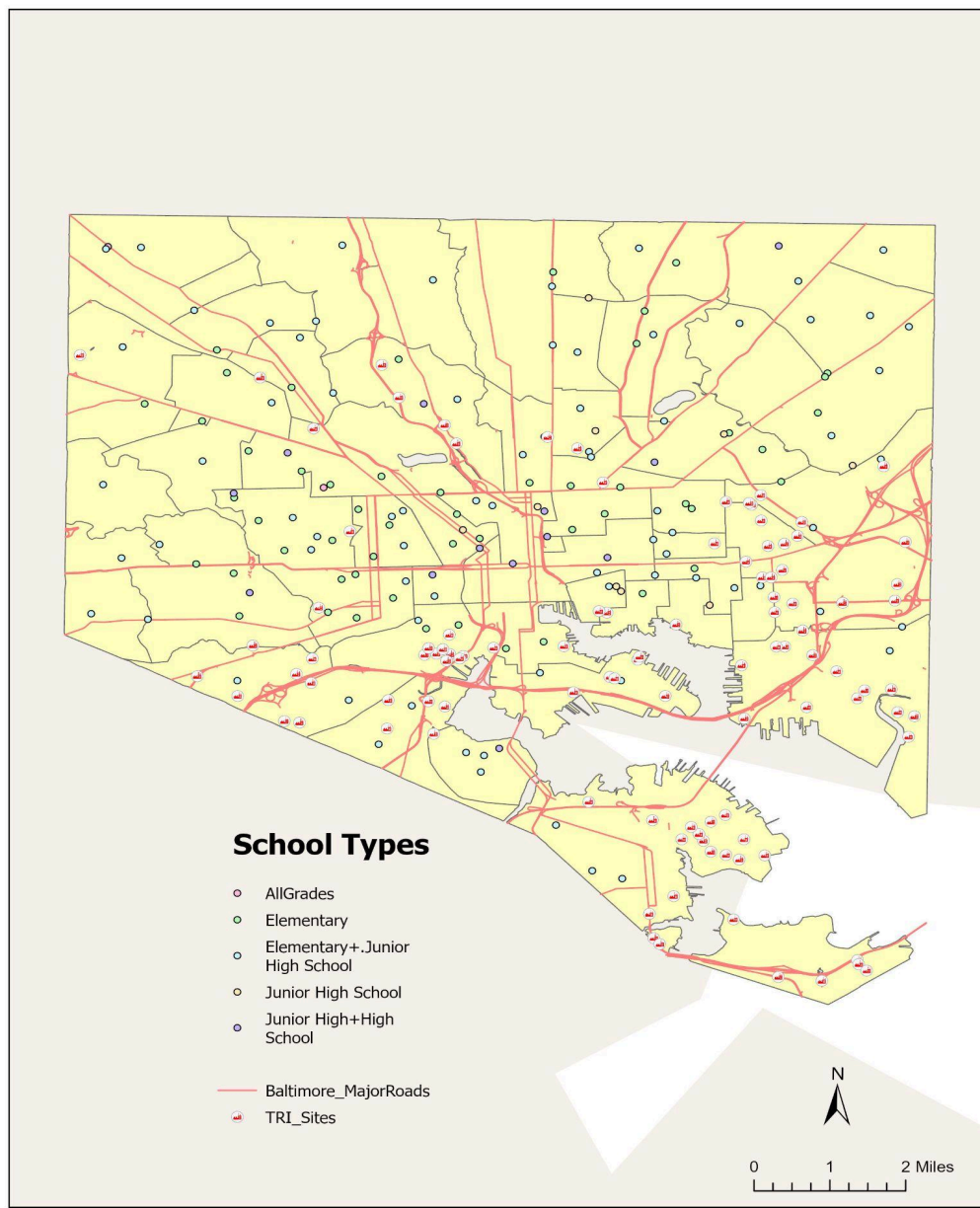


Thanh Doan
PUBH6141
Assignment 1

Question 1. Produce a map similar to Figure 1 below. Your map is to include all the elements of a good map (including an indicator map). Your map must show 1) Baltimore City schools, 2) stratification of schools by type, 3) major roadways, and 4) toxic release inventory (TRI) sites. Feel free to use your own color and shape schemes, but your map must be visually pleasing. Remember, your map is to emphasize Baltimore School locations (35 points)



Question 2. Using your public health knowledge, describe why TRI sites and major roads might be important when evaluating school performance and absenteeism? (5 points)

The location of both TRI sites and major roadways is important for evaluating both performance and absenteeism. If we find that the locations of TRI sites are close to the locations of schools with lower performance or higher absenteeism, it may mean that the students are being affected by the toxic environment, leading to learning disabilities or illness. Major roadways could have an impact on accessibility to education (i.e. schools). If it is difficult for students to get to school, either by difficulty going to school by car or lack of bus routes near their home, their performance and absenteeism may be affected.

Question 3. Consider other environmental hazards that might be important when assessing the health of school children. Discuss one new piece of spatial/geographic data you would like to add to your map (you do not actually have to find this data)? Briefly describe this potential data, its spatial data type (remember there are 3 types of spatial data), and how you might display it using symbology or colors. (5 points)

I feel that another environmental hazard that should be looked at would be manufacturing plants that are known to create pollution. Similar to the TRI plants, manufacturing plants create far reaching pollutants in the form of carbon dioxide from burning fuel, to chemicals created in the manufacturing process. These are often released into the air and can reach several miles outward from the plant, causing a toxic environment.

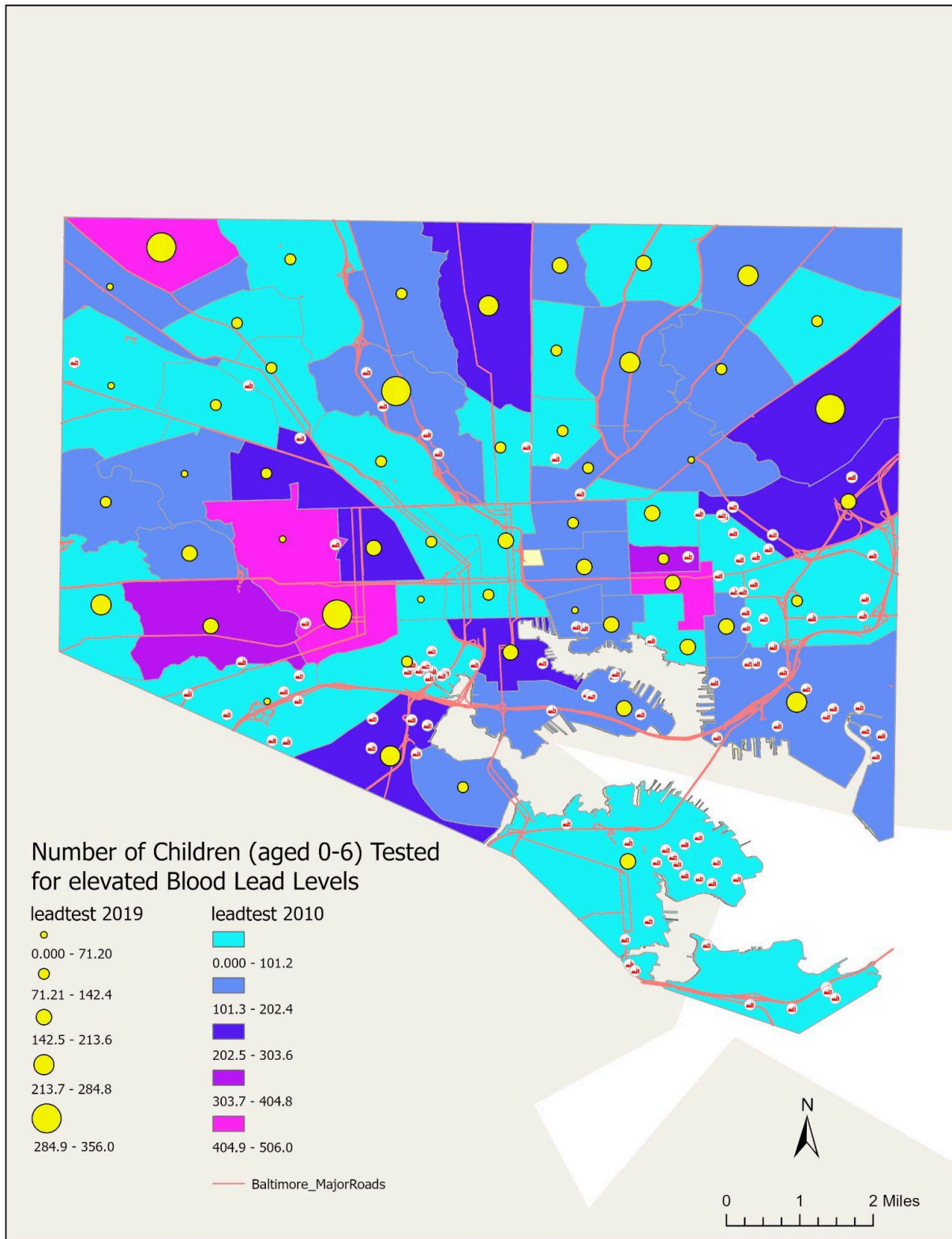
This would be point pattern data that I would represent with the icon for an industrial complex in a bright neon green to denote toxicity.

Question 4. Produce a single map of the Baltimore City CSA's displaying TWO of the new variables you downloaded. (35 points)

- Note: There is no need to display schools, TRI sites, or major roads. However, you may include these if you believe important for your map and/or answering questions 5 and 6

- Note: This map does not need an indicator map, but should include a legend, scale bar, and north arrow

- Hint: Think about using two forms of symbology for your two layers. For example, color and shape size.

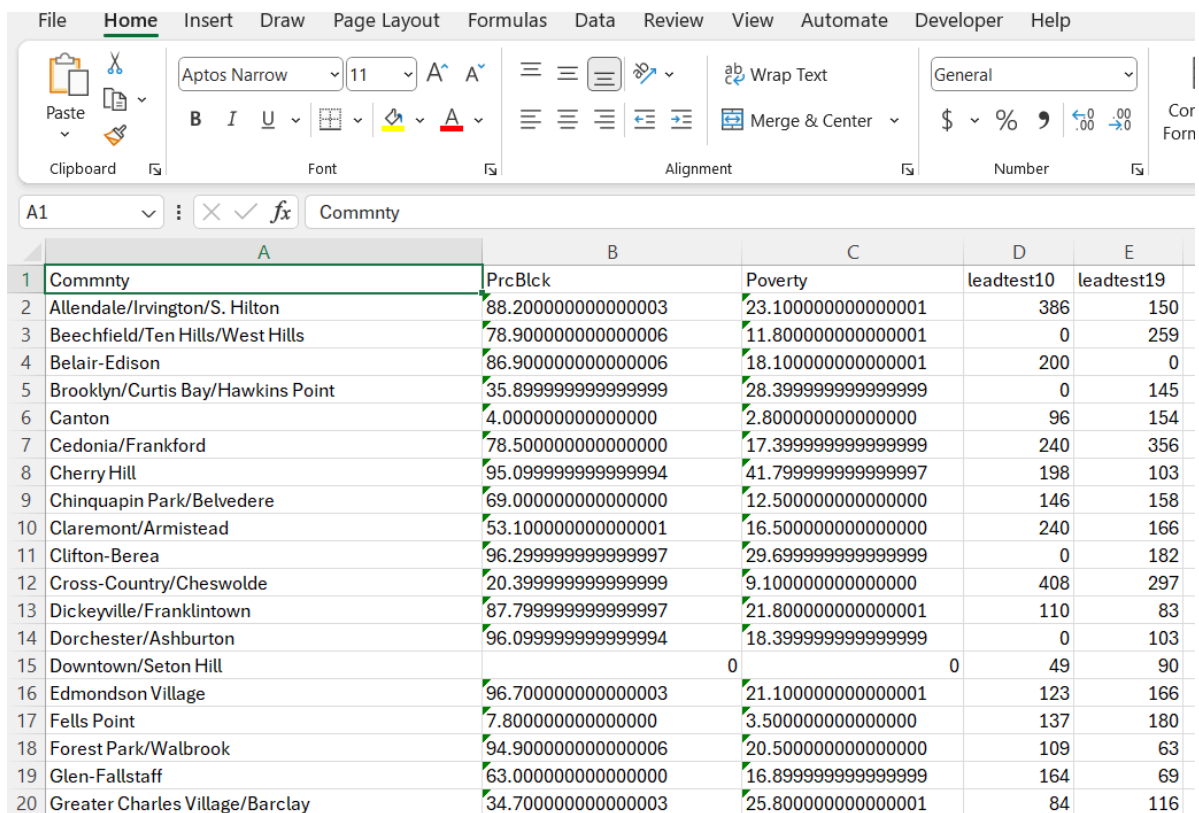


Question 5. Write a short paragraph discussing your new map. Clearly state what it displays and the units. Explain why you selected a particular color scheme, symbology, or category to map. Then describe how your map might be used in a health effects study. (15 points)

- Note: You may write this up in paragraph or bulleted form

For the new map, I used the file 'Number of Children (aged 0-6) Tested for Elevated Blood Lead Levels - Community Statistical Area' found on Vital Signs. For this map, I displayed the number of children tested for elevated blood levels for the years 2010 and 2019. The data for 2010 is displayed via a graduated colors scheme that colors the areas according to the number of children tested. The data for 2019 is displayed via a graduated symbols scheme that increases in size according to the number of children tested. I kept the TSI data and the data for the main roads to see if there is a connection between the location of the TSI sites and the elevated number of children tested for lead.

Question 6. Join your new data with the original CSA shapefile. Reduce your attribute table to show the Community name, 2 original CSA variables, and your 2 variables shown in Question 4. Export to a CSV file or .txt file and open in Excel. Paste a screenshot showing the data table opened in Excel. (5 points)



	A	B	C	D	E
1	Commnty	PrcBlck	Poverty	leadtest10	leadtest19
2	Allendale/Irvington/S. Hilton	88.200000000000003	23.100000000000001	386	150
3	Beechfield/Ten Hills/West Hills	78.900000000000006	11.800000000000001	0	259
4	Belair-Edison	86.900000000000006	18.100000000000001	200	0
5	Brooklyn/Curtis Bay/Hawkins Point	35.899999999999999	28.399999999999999	0	145
6	Canton	4.000000000000000	2.800000000000000	96	154
7	Cedonia/Frankford	78.500000000000000	17.399999999999999	240	356
8	Cherry Hill	95.099999999999994	41.799999999999997	198	103
9	Chinquapin Park/Belvedere	69.000000000000000	12.500000000000000	146	158
10	Claremont/Armistead	53.100000000000001	16.500000000000000	240	166
11	Clifton-Berea	96.299999999999997	29.699999999999999	0	182
12	Cross-Country/Cheswolde	20.399999999999999	9.100000000000000	408	297
13	Dickeyville/Franklinton	87.799999999999997	21.800000000000001	110	83
14	Dorchester/Ashburton	96.099999999999994	18.399999999999999	0	103
15	Downtown/Seton Hill	0	0	49	90
16	Edmondson Village	96.700000000000003	21.100000000000001	123	166
17	Fells Point	7.800000000000000	3.500000000000000	137	180
18	Forest Park/Walbrook	94.900000000000006	20.500000000000000	109	63
19	Glen-Fallstaff	63.000000000000000	16.899999999999999	164	69
20	Greater Charles Village/Barclay	34.700000000000003	25.800000000000001	84	116